Bachelor of Vocational (B.Voc). (Medical Lab Technology)
Job Role – Medical Lab Technician

SEMESTER I

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN - 101	Communication Skills	Generic	Theory	20	80	--	--	6
*GEN-102	Fundamentals of Information Technology	Generic	Theory	20	80	--	--	6
MLT103	Basics Of Human Anatomy	Skill	Theory & Practical	10	40	10	40	6
MLT104	Introduction To Laboratory Equipments	Skill	Theory & Practical	10	40	10	40	6
MLT105	Introduction To Hematology	Skill	Theory & Practical	10	40	10	40	6

SEMESTER II

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN - 201	Soft Skills and Personality Development	Generic	Theory	20	80	--	--	6
MLT 107	Introduction To Medical Technology Healthcare Systems	Generic	Theory & Practical	10	40	10	40	6
MLT108	Basics Of Phsiology	Skill	Theory & Practical	10	40	10	40	6
MLT109	Clinical Biochemistry	Skill	Theory & Practical	10	40	10	40	6
MLT110	Analytical Laboratory Testing Process -I	Skill	Theory & Practical	10	40	10	40	6
	Industrial Visit/Training Report And Making And Making							NC**

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

**** Non Credit**

Job Role – Medical Lab Technician

SEMESTER I
MLT 103

BASICS OF HUMAN ANATOMY

Objectives: Basic understanding of organization of body cells, tissues, organs, organ systems, and glands in human body

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Section I

- Basic unit of body - Cell
- The anatomical organization of human body
- Introduction to different types of tissues: Anatomy, epithelial tissue, muscular tissue, nervous tissue. Different types of organ systems.
- Skeletal system: bones, joints and muscles.

Section II

Digestive Organs:
- Tongue
- Teeth
- Oral cavity
- Pharynx
- Oesophagus
- Stomach
- Small intestine
- Large intestine and its colons
- Liver, Pancreas and Spleen

Section III

Respiratory Organs:
- Nasopharynx
- Oropharynx
- Larynx

- Trachea
- Bronchi
- Lungs
- Thoracic cavity
- Pleura and Pleural cavity

Circulation System:
Structure of heart, names and position of main blood vessels.

Section IV

Reproductive Organs Male and Female Gonads

E) Excretory Organs:
- Cortex and medulla of Kidney
- Ureter
- Urinary Bladder

Urethra (male and female)

PRACTICAL: Study of various organs through Charts and models

Reference Books:

Anatomy & Physiology	Ross and Wilson
Anatomy and Physiology	N Murgesh
Anatomy and Physiology for nurses	Evelyn Pearce
Anatomy and Physiology for nurses	Sears
Anatomy and Physiology for nurses	Pearson
Human Anatomy,	Harie R. Berasari

MLT 104
INTRODUCTION TO LABORATORY EQUIPMENTS

Objectives:

- To gain broad understanding of care of laboratory glassware, equipment and instrument
- To gain broad understanding of setting up, calibrating, operating, cleaning, maintaining, troubleshooting of laboratory equipment used in quantitative or qualitative analysis
- To Calibrate and Validate the Clinical Laboratory instruments and glass wares
- To understand Microscopy, working principle, maintenance and applications of various types of microscopes

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Section I

Introduction to operation and safety precautions of common Laboratory Equipments Principles and working of laboratory instruments Incubator, Hot Air Oven, Water Bath , Anaerobic Jar, Centrifuge, Autoclave, burettes & pipettes ,colorimeter neubauer's Chamber Setting up, calibrating and validating, operating, cleaning, maintaining, troubleshooting of laboratory equipment used in quantitative or qualitative analysis Autoclave – its structure, functioning, control and indicator Micrometry

Section II

- Techniques of Disinfection
- Sterilization : Definition, Classification and General Principle of Sterilization, Sterilization Techniques, Sterilization of rubber goods, laboratory equipment & other instruments
- The cleaning and maintenance procedures of the machine
- Identify the cause of errors or other problems or defects in equipments
- Glassware – Description of Glassware, its use, handling and care
- Importance and methods of cleaning of glass apparatus
- Calibration of apparatus and glasswares

Section III

- To study the compound microscope and its parts
- Microscopy, working principle, maintenance and applications of various types of microscopes:-

- Dark ground microscope.
- Polarizing microscope.
- Phase contrast microscope.
- Interference microscope.
- U.V. light microscope.

SECTION IV

- Microscopic examination of micro-organism, bright field microscopy, dark field microscopy, phase contrast microscopy, electron microscopy.

- Electron microscope: working principle, components and allied techniques for electron microscopy, ultra-microtomy Care and handling of various microscopes-Binocular, DGI, Phase-contrast, fluorescence and electron microscopes
- Museum techniques
- Lab safety and Instrumentation

Reference Books

At the Bench : A Laboratory Navigator	Kathe Barker
At the Helm : A Laboratory Navigator	Kathe Barker
Basic Medical Laboratory techniques	Barbara H. Estridge *et al*
Instrumental Analysis	Chatwal Anand
Laboratory Reference	Jane Roskams
Medical Dictionary	Oxford
Medical Informatics	Mohan Bansal

MLT 105
INTRODUCTION TO HEMATOLOGY

Objectives:

- **To gain understanding of blood and components of blood**
- **To gain knowledge of hematological Diseases and hematological Investigations.**

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Section I

- Introduction to hematology and laboratory organization
- Formation of blood.
- Composition and functions of blood. The chemical composition, structure, and properties of substances, Morphology of normal blood cells and their identifications
- Heamostasis ,Mechanism of blood coagulation. Fibrinolysis. , body fluids.

Section II

- Various anticoagulants, their uses, mode of action and their merits and demerits.
- Normal and absolute in haematology.
- Quality assurance in hematology

Section III

- Descriptive study of RBC abnormalities, Disorders related to RBC Normal white cell count & physiological variation Normal white cell count & physiological variation
- **Hematological Diseases:** Anemia and various types of anemia, Thalassemia, Polycythemia, hemolytic disease of new born, multiple myeloma, parasitic infections of blood
- Leukemia: definition and classification (General & FAB)

Section IV

Laboratory Investigation & Bleeding Disorders

- Laboratory preparation for coagulation tests
- Routine coagulation tests, (prothrombin time , plasma recalcification time ,partial thromboplastin time , activated partial thromboplastin time, thrombin time, Laboratory diagnosis of bleeding disorders .

PRACTICALS

- Cleaning of Laboratory glassware in hematology
- Clinical significance ,specimen collection, laboratory investigation & preservation of blood for various hematological investigations

- Preparation of blood smear.
- Total leukocyte count and Differential leukocyte count.
- Haemoglobinometery, various methods of estimation of Hb, errors involved and standardization of instrument for adaptation for Hb estimation.
- Haemocytometery, procedures for cell counts-visual as well as electronic.
- Determination of total erythrocyte(RBC) count and platelet counts. Errors involved and mean to minimize such errors.
- Romanowskydyes , preparation and staining procedures of blood smears.
- Laboratory tests for assessing bleeding disorders

Reference Books

Atlas of haematology (5/e)	G.A. McDonald
Clinical Haematology	Christopher A. Ludlam
Practical Haematology	J.B. Dacie
Practical Haematology (8/e) S	ir John
Haematology (International edition)	Emmanuel C.Besa
Haematology (Pathophysiological basis for clinical practice (3/e)	Stephen M. Robinson
Haematology for students Practitioners	Ramnik Sood
Hand book of Medical Laboratory Technology (2/e)	V.H. Talib

SEMESTER II

MLT 107 INTRODUCTION TO MEDICAL TECHNOLOGY HEALTHCARE SYSTEMS

Objective:

- To Understand about Healthcare Service Providers
- To develop broad understanding of the Role of MLT
- To Understand Patient's Rights &Responsibilities

Section I

- Introduction to medical technology
- Healthcare Systems, Laboratory and Delivery
- Role of Medical Laboratory Technician
- Patient Comforts and Safety
- Understanding of Laboratory Test Results
- Ethical Behavior

Section II

- Patient's Rights &Responsibilities
- Understand sensitivities involved in patient's right
- Learn medical laboratory technician's role in maintaining patient's rights
- Patient's Environment

SECTION III

- Use of laboratory related medical terminology in daily activities with colleagues, patients and family
- Monitor and assure quality and quality assurance program
- The organisations' policies and commitments towards quality assurance

SECTION IV

- Maintain a safe, healthy, and secure working environment
- The importance of health, safety, and security in the workplace
- Common Hazard
- Create safety records and maintaining them
- Organisational structure and the various processes related to reporting and monitoring
- Procedure for accessing training, learning and development needs

Reference Books

A Manual of Laboratory & Diagnostic Tests (6/ e)	Frances Fischbach
Hand book of Medical Laboratory Technology (2/e)	V.H. Talib
Clinical Diagnosis & Management by Laboratory method0 (20/e)	John Bernard Henary
Textbook of Medical Laboratory Technology	Godkar and Godkar

MLT 108
BASICS OF PHSIOLOGY

Objectives:

- Basic understanding of physiology of different organ system of body.

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

SECTION I

- Nutrition (Vitamins, Calories)
- Integumentary system
- Digestive system, mechanism and physiology of digestion and absorption.

SECTION II

- Lymphatic system, different types of body fluids: secretions and excretions,
- Spleen, lymph node and R.E. system
- Cardiovascular system and blood vessels

SECTION III

- Respiratory system.
- Musculo-skeletal system
- Reproductive System

SECTION IV

- Nervous system and Sense organs
- Endocrine glands and their functions.

PRACTICAL

- Study of systems mentioned in theory through Charts.
- To separate the plasma and serum from given blood sample
- Study of various endocrine glands through permanent slides

Reference books

Textbook of Medical Physiology	Guyton and Hall
Anatomy & Physiology	Ross and Wilson
Anatomy and Physiology	N Murgesh
Anatomy and Physiology for nurses	Evelyn Pearce
Anatomy and Physiology for nurses	Sears
Anatomy and Physiology for nurses	Pearson
Anatomy and Physiology: Understanding the Human Body	Clark
Physiology & Health Education	Gandhi & Goel
Endocrinology	Headley
Human Physiology	Andrew Davis
Manual of Endocrinology and Metabolism	Norman Levin

MLT 109
Clinical Biochemistry

Objectives:

- Elementary knowledge of Carbohydrates, lipids, Proteins, Enzymes,Clinical enzymology.
- Elementary knowledge of Hormones
- Elementary knowledge of Minerals and Electrolytes
- To Understand about Therapeutic Drug Monitoring

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

Section I

- Introduction and general properties of Nucleic acids and Enzymes.
- Clinical enzymology

Section II

Enzyme Histochemistry and Estimation of important Amino acids and enzymes: Enzymes Acid Phosphatase (ACP), Alkaline Phosphatase (ALP), Transaminase, LDH, AST, ALT, Amylase and lactate dehydrogenase, Creatinine, Creatine Phosphokinase (CPK) , CPK-MB.

Determination of serum glutamate pyruvate transaminase (SGPT) and serum glutamate Oxaloacetate transaminase (SGOT) End point reaction

Section III

- Elementary knowledge of Minerals and Electrolytes
- Therapeutic Drug Monitoring

Section IV

Liver tests, Renal tests, Endocrine function tests, Lipid profile

Practical

Liver tests, Renal tests, Endocrine function tests, Lipid profile,
Estimation of important Amino acids and enzymes: Enzymes Acid Phosphatase (ACP), Alkaline Phosphatase (ALP), Transaminase, LDH, AST, ALT, Amylase and lactate dehydrogenase, Creatinine, Creatine Phosphokinase (CPK) , CPK-MB.
Determination of serum glutamate pyruvate transaminase (SGPT) and serum glutamate Oxaloacetate transaminase (SGOT) End point reaction

Reference Books

A guidebook to Biochemistry	Michael Yudkin
A Manual of Laboratory & Diagnostic Tests (6/ e)	Frances Fischbach
Biochemistry	Voet and Voet
Biochemistry	Stryer
Biochemistry	U. Satyanarayan. & U. Chakrapani
Clinical Biochemistry	Richard Luxton
Clinical Diagnosis & Management by Laboratory method0 (20/e)	John Bernard Henary
Clinical Biochemistry	G. Guru
Handbook of Biochemistry	M.A. Siddique
Textbook of Medical Biochemistry	S. Ramkrishnan
Biochemical Techniques	K. Choudhary
Text book of Medical Biochemistry	Chaterjee & Shinde
Principles of Biochemistry	David L. Nelson
Principles of Biochemistry	Lehninger
Textbook of Biochemistry and Human Biology	G.P. Talwar
Textbook of Medical Laboratory Technology	Godkar and Godkar
Outline of Biochemistry	Conn Stumpf

Principles of Internal Medicine	Isselbacher
Proteins and Proteomics : Laboratory Manual	Richard J. Simpson
Purifying Proteins for Proteomics: Laboratory Manual	Richard J. Simpson

MLT 110
Analytical Laboratory Testing Process-I

Objectives:

- To gain broad understanding of chemicals/reagents useful in sample analysis.
- To gain broad knowledge of Routine Hematological Tests and Urine tests,Stool tests, Semen tests and sputum tests

Instructions:

- The syllabus of this paper has been divided into four units.
- Examiner will set a total of nine questions comprising two questions from each unit,
- Question number **one** is compulsory of short answer type questions covering the whole syllabus.
- The students are required to attempt one question from each unit.
- All questions carry equal marks

SECTION I

- Collection and recording of biological specimens
- Collection and Handling of Blood Sample

Routine Hematological Tests

- Examinations of sample for serum Urea and Uric acid
- Estimation of Essential electrolytes: Sodium, potassium, calcium, chloride and phosphorus etc.
- Determination of blood sugar level of plasma (or serum) (a) Orthotoluidine method, (b) Glucose oxidase method
- Determination od the serum urea nitrogen (a) Diacetyl monoxime method
- Determination of serum creatinine : Alkaline picrate method
- Determination of serum total cholesterol

SECTION II

- Determination of serum bilirubin a.Malloy and Evelyn b.DMSO method
- Determination of TSH
- Laboratory investigation for megaloblastic anaemia
- Laboratory investigation for iron deficiency Anaemia
- Laboratory investigation for haemolytic anaemia including classification and causes
- Laboratory investigation for disseminated intravascular coagulation(DIC)

- Mechanism of fibrinolysis test for fibrinolysis
- Platelet function tests and their interpretation

SECTION III

- Erythrocyte sedimentation rate, factors influencing ESR and various procedures for its estimation with their significance.
- Haematocrit value by macro and micro methods their merits and demerits.
- Determination of pack cell volume (PCV)
- Physiological variations in HB, PCV, TLC and platelets
- Determination and calculation of red blood indices MCH,MCH,MCHC
- Determination of absolute Eosinophil count and Reticulocyte count
- Determination of hematocrit,
- Enumeration of formed elements,
- Automated systems in hematology

SECTION IV

Urine Examination

- Collection, Handling, Examination of Urine
- Urine analysis, routine examination of urine(physical examination of urine) rapid chemical tests of Urine, Determination of specific gravity of urine by urinometer and refractormeter, Microscopic examination of urine
- Clinical significance and Detailed Examination of Urine

Biochemical Test Profile (Quantitative determination of Urine)

Amylase, Calcium, Chlorides, Creatinine, Sodium, Potassium, Glucose, Proteins, Urea nitrogen, uric acid bile pigments, ketone bodies, porphobilinogen, faccal occult blood.

Reference Books

Textbook of Medical Laboratory Technology	Godkar and Godkar
Research Methodology in Medical Sciences	Chandorkar
Practical Clinical Biochemistry	Harold Varley
Medical Laboratory Sciences, Theory & Practical	A. Kolhatkar
Medical Laboratory Technology – Volume I	Kanai Mukherjee
Medical Laboratory Technology – Volume II	Kanai Mukherjee
Medical Laboratory Technology – Volume II	Kanai Mukherjee
Medical Laboratory Technology Methods & Interpretation (5/e)	Ramnik Sood

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